(a) Using the substitution $u = \sqrt{2x+1}$, show that

$$\int_4^{12} \sqrt{8x+4}\ \mathrm{e}^{\sqrt{2x+1}}\ \mathrm{d}x$$

may be expressed in the form

$$\int_a^b ku^2\mathrm{e}^u\ \mathrm{d}u$$

where a, b and k are constants to be found.

(4)

(b) Hence find, by algebraic integration, the exact value of

$$\int_4^{12} \sqrt{8x+4}\ \mathrm{e}^{\sqrt{2x+1}}\ \mathrm{d}x$$

giving your answer in simplest form.

(5)